STAT 339 METHODS OF LINEAR ALGEBRA
Inner Product Spaces

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Outline

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Length and Dot Product in R^n

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Length

The length of a vector $\mathbf{v} = (v_1, v_2, \cdots, v_n)$ in R^n is given by

$$||\mathbf{v}|| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}, \qquad (||\mathbf{v}|| \text{ is a real number})$$

Properties of length (or norm)

1. $||\mathbf{v}|| \geq 0$
2. $||\mathbf{v}|| = 1 \implies v$ is all a unit vector
3. $||\mathbf{v}|| = 0$ if and only if $\mathbf{v} = \mathbf{0}$
4. $||c\mathbf{v}|| = |c| \, ||\mathbf{v}||$

Note

The length of a vector is also called its norm

Example

1. In R^5, the length of $\mathbf{v} = (0, -2, 1, 4, -2)$ is given by

$$||\mathbf{v}|| = \sqrt{(0)^2 + (-2)^2 + (1)^2 + (4)^2 + (-2)^2} = \sqrt{25} = 5$$

2. In R^3, the length of $\mathbf{v} = (\frac{2}{\sqrt{17}}, \frac{-2}{\sqrt{17}}, \frac{3}{\sqrt{17}})$ is given by

$$||\mathbf{v}|| = \sqrt{\left(\frac{2}{\sqrt{17}}\right)^2 + \left(\frac{-2}{\sqrt{17}}\right)^2 + \left(\frac{3}{\sqrt{17}}\right)^2} = \sqrt{\left(\frac{17}{17}\right)} = 1$$

Example (A standard unit vector in R^n: only one component of the vector is 1 and the others are 0 (thus the length of this vector must be 1))

1. $R^2 : \{\mathbf{e}_1, \mathbf{e}_2\} = \{(1,0),(0,1)\}$,
2. $R^3 : \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \{(1,0,0),(0,1,0),(0,0,1)\}$,
3. $R^n : \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \cdots, \mathbf{e}_n\} = \{(1,0,\cdots,0),(0,1,\cdots,0),(0,0,\cdots,1)\}$,

Note

Two nonzero vectors are parallel if $\mathbf{u} = c\mathbf{v}$

1. $c > 0 \implies$ **u** and **v** have the same direction
2. $c < 0 \implies$ **u** and **v** have the opposite directions

Theorem ((5.1) Length of a scalar multiple)

Let $\mathbf{v}$ *be a vector in* R^n *and* c *be a scalar. Then*

$$||c\mathbf{v}|| = |c|\,||\mathbf{v}||$$

Proof.

$$\begin{aligned}
\mathbf{v} =& (v_1, v_2, \cdots, v_n)\\
\implies c\mathbf{v} =& (cv_1, cv_2, \cdots, cv_n)\\
||c\mathbf{v}|| =& ||(cv_1, cv_2, \cdots, cv_n)||\\
=& \sqrt{(cv_1)^2 + (cv_2)^2, \cdots, (cv_n)^2} = \sqrt{c^2((v_1)^2 + (v_2)^2 + \cdots, (v_n)^2)}\\
=& |c|\sqrt{(v_1)^2 + (v_2)^2 + \cdots + (v_n)^2} = |c|\,||\mathbf{v}||
\end{aligned}$$

Theorem ((5.1) Length of a scalar multiple)

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$$\begin{aligned}
\mathbf{v} =& (v_1, v_2, \cdots, v_n) \\
\implies c\mathbf{v} =& (cv_1, cv_2, \cdots, cv_n) \\
||c\mathbf{v}|| =& ||(cv_1, cv_2, \cdots, cv_n)|| \\
=& \sqrt{(cv_1)^2 + (cv_2)^2, \cdots, (cv_n)^2} = \sqrt{c^2((v_1)^2 + (v_2)^2 + \cdots, (v_n)^2)} \\
=& |c|\sqrt{(v_1)^2 + (v_2)^2 + \cdots + (v_n)^2} = |c|\,||\mathbf{v}||
\end{aligned}$$

Theorem ((5.2) How to find the unit vector in the direction of $\mathbf{v}$)

If $\mathbf{v}$ is a nonzero vector in R^n then the vector $\mathbf{u} = \frac{\mathbf{v}}{||\mathbf{v}||}$ has has length 1 and has the same direction as $\mathbf{v}$. This vector $\mathbf{u}$ is called the unit vector in the direction of $\mathbf{v}$

Proof.

$$\mathbf{v} \text{ is nonzero} \implies ||\mathbf{v}|| \leq 0 \implies \frac{1}{||\mathbf{v}||} \leq 0$$

$$\text{If } \mathbf{u} = \frac{1}{||\mathbf{v}||}\mathbf{v} \text{ } (\mathbf{u} \text{ has the same direction as v})$$

$$||\mathbf{u}|| = \left|\left|\frac{\mathbf{v}}{||\mathbf{v}||}\right|\right| = \frac{1}{||\mathbf{v}||}||\mathbf{v}|| = 1$$

Theorem ((5.2) How to find the unit vector in the direction of $\mathbf{v}$)

If $\mathbf{v}$ is a nonzero vector in R^n then the vector $\mathbf{u} = \frac{\mathbf{v}}{||\mathbf{v}||}$ has has length 1 and has the same direction as $\mathbf{v}$. This vector $\mathbf{u}$ is called the unit vector in the direction of $\mathbf{v}$

Proof.

$$\mathbf{v} \text{ is nonzero} \implies ||\mathbf{v}|| \leq 0 \implies \frac{\mathbf{1}}{||\mathbf{v}||} \leq 0$$

$$\text{If } \mathbf{u} = \frac{\mathbf{1}}{||\mathbf{v}||}\mathbf{v} \text{ } (\mathbf{u} \text{ has the same direction as v})$$

$$||\mathbf{u}|| = \left|\left|\frac{\mathbf{v}}{||\mathbf{v}||}\right|\right| = \frac{\mathbf{1}}{||\mathbf{v}||}||\mathbf{v}|| = 1$$

$\square$

Note

- The vector $\frac{\mathbf{v}}{||\mathbf{v}||}$ is called the unit vector in the direction of $\mathbf{v}$
- The process of finding the unit vector in the direction of $\mathbf{v}$ is called normalizing the vector $\mathbf{v}$

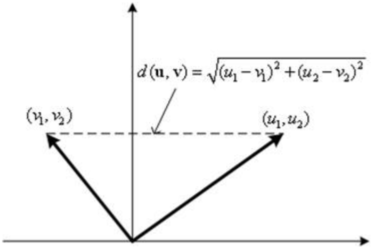
Example (Finding a unit vector)

Find the unit vector in the direction of $\mathbf{v} = (3, -1, 2)$, and verify that this vector has length 1.

Distance between two vectors

The distance between two vectors **u** and **v** in R^n is given by

$$d(\mathbf{u}, \mathbf{v}) = ||\mathbf{u} - \mathbf{v}||$$



Properties of length (or norm)

1. $d(\mathbf{u}, \mathbf{v}) = \geq 0$
2. $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$
3. $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u}) =$ (commutative property of the distance function)

Example (Finding the distance between two vectors)

The distance between $\mathbf{u} = (0, 2, 2)$ and $\mathbf{v} = (2, 0, 1)$ is

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}) &= ||\mathbf{u} - \mathbf{v}|| = |(0-2, 2-0, 2-1)|| \\ &= \sqrt{(-2)^2 + (2)^2 + (1)^2} = 3 \end{aligned}$$

Dot Product in R^n

The dot product of $\mathbf{u} = (u_1, u_2, \cdots, u_n)$ and $\mathbf{v} = (v_1, v_2, \cdots, v_n)$ returns a scalar quantity

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

(The dot product is defined as the sum of component-by-component multiplications)

Theorem ((5.3) Properties of the dot product)

If $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ are vectors in R^n and c is a scalar, then the following properties are true

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ *(commutative property of the dot product)*
2. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ *(distributive property of the dot product over vector addition)*
3. $c(\mathbf{u} \cdot \mathbf{v}) = c(\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v})$ *(associative property of the scalar multi. and the dot product)*
4. $\mathbf{v} \cdot \mathbf{v} = ||\mathbf{v}||^2 \implies \mathbf{v} \cdot \mathbf{v} \geq 0$
5. $\mathbf{v} \cdot \mathbf{v} = 0$ *if and only* $\mathbf{v} = \mathbf{0}$

Example (Finding the dot product of two vectors)

The dot product of $\mathbf{u} = (1, 2, 3, -3)$ and $\mathbf{v} = (3, -2, 4, 2)$ is

$$\begin{aligned}\mathbf{u} \cdot \mathbf{v} =& (1)3 + (2)(-2) + (0)(4) + (-3)(2) \\ =& -7\end{aligned}$$

Length and Dot Product in R^n

Euclidean n-space

- In week 4, R^n was defined to be the set of all order n-tuples of real numbers
- When R^n is combined with the standard operations of vector addition, scalar multiplication, vector length, and dot product, the resulting vector space is called Euclidean n-space

Example (Find dot products)

Consider $\mathbf{u} = (2, -2)$, and $\mathbf{v} = (5, 8)$ and $\mathbf{w} = (-4, 3)$. Find the following

1. $\mathbf{u} \cdot \mathbf{v}$
2. $(\mathbf{u} \cdot \mathbf{v})\mathbf{w}$
3. $\mathbf{u} \cdot (2\mathbf{v})$
4. $||\mathbf{w}||^2$
5. $\mathbf{u} \cdot (\mathbf{v} - 2\mathbf{w})$

Example (Using the properties of the dot product)

Given $\mathbf{u} \cdot \mathbf{u} = 39$, and $\mathbf{u} \cdot \mathbf{v} = -3$ and $\mathbf{v} \cdot \mathbf{v} = 79$. Find $(\mathbf{u} + 2\mathbf{v}) \cdot (3\mathbf{u} + \mathbf{v})$

Theorem ((5.4) The Cauchy-Schwarz inequality)

If $\mathbf{u}$ *and* $\mathbf{v}$ *are vector in* R^n *then*

$$|\mathbf{u}\cdot\mathbf{v}| \leq ||\mathbf{u}||\,||\mathbf{v}||, \qquad (|\mathbf{u}\cdot\mathbf{v}| \text{ denotes the absolute value of } |\mathbf{u}\cdot\mathbf{v}|)$$

(The geometric interpretation for this inequality is shown on the next slide)

Example (An example of the Cauchy-Schwarz inequality)

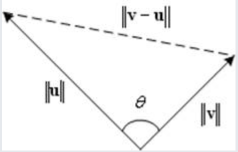
Verify the Cauchy-Schwarz inequality for $\mathbf{u} = (1, -1, 3)$, and $\mathbf{v} = (2, 0, -1)$.

Length and Dot Product in R^n

Dot product and the angle between two vectors

To find the angle $\theta (0 \leq \theta \leq \pi)$ between two nonzero vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ in R^2, the Law of Cosines can be applied to the following triangle to obtain

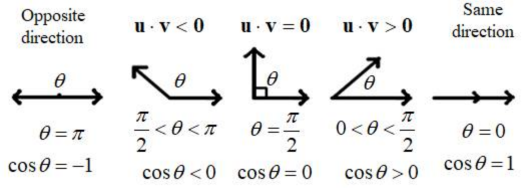
$$||\mathbf{v} - \mathbf{u}||^2 = ||\mathbf{v}||^2 + ||\mathbf{u}||^2 - 2||\mathbf{v}||||\mathbf{u}|| \cos\theta$$



$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||||\mathbf{u}||}, 0 \leq \theta \leq \pi$$

The angle between two nonzero vectors in R^n

$$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{v}||||\mathbf{u}||}, 0 \leq \theta \leq \pi$$



Note

The angle between the zero vector and another vector is not defined (since the denominator cannot be zero)

Example (Finding the angle between two vectors)

Consider $\mathbf{u} = (-4, 0, 2, -2)$, and $\mathbf{v} = (2, 0, -1, 1)$.

Solution

$$
\begin{aligned}
||\mathbf{u}|| =& \sqrt{\mathbf{u}\cdot\mathbf{u}} = \sqrt{(-4)^2 + 0^2 + 2^2 + (-2)^2} = \sqrt{24} \\
||\mathbf{v}|| =& \sqrt{\mathbf{v}\cdot\mathbf{v}} = \sqrt{2^2 + 0^2 + (-1)^2 + 1^2} = \sqrt{6} \\
\mathbf{u}\cdot\mathbf{v} =& (-4)(2) + (0)(0) + (2)(-1) + (-2)(1) = -12 \\
\implies \cos\theta =& \frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{v}||||\mathbf{u}||} = \frac{-12}{\sqrt{24}\sqrt{6}} = -\frac{12}{144} = -1
\end{aligned}
$$

$\implies \theta = \pi \therefore \mathbf{u}$ *and* $\mathbf{v}$ *have opposite directions*

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Orthogonal vectors

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in R^n are orthogonal (perpendicular) if

$$\mathbf{u} \cdot \mathbf{v} = 0$$

Note

The vector $\mathbf{0}$ is said to be orthogonal to every vector

Length and Dot Product in R^n

Example (Finding orthogonal vectors)

Determine all vectors in R^n that are orthogonal to $\mathbf{u} = (4, 2)$.

Solution

Given $\mathbf{u} = (4, 2)$, *let* $\mathbf{v} = (v_1, v_2)$

$$\begin{aligned}
\implies \quad \mathbf{u} \cdot \mathbf{v} =& (4, 2) \cdot (v_1, v_2) \\
=& 4v_1 + 2v_2 = 0 \\
\implies \quad v_1 =& \frac{-t}{2}, v_2 = t \\
\therefore \mathbf{v} =& \left(\frac{-t}{2}, t \right), t \in \Re
\end{aligned}$$

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Length and Dot Product in R^n

Theorem ((5.5) The triangle inequality)

If $\mathbf{u}$ *and* $\mathbf{v}$ *are vector in* R^n*, then* $||\mathbf{u}+\mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$.

Proof.

$$
\begin{aligned}
||\mathbf{u}+\mathbf{v}||^2 &= (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{v}) \\
&= \mathbf{u}\cdot(\mathbf{u}+\mathbf{v}) + \mathbf{v}\cdot(\mathbf{u}+\mathbf{v}) = \mathbf{u}\cdot\mathbf{u} + 2(\mathbf{u}\cdot\mathbf{v}) + \mathbf{v}\cdot\mathbf{v} \\
&= ||\mathbf{u}||^2 + 2(\mathbf{u}\cdot\mathbf{v}) + ||\mathbf{v}||^2 \leq ||\mathbf{u}||^2 + 2|\mathbf{u}\cdot\mathbf{v}| + ||\mathbf{v}||^2 (c \leq |c|) \\
&\leq ||\mathbf{u}||^2 + 2||\mathbf{u}||\,||\mathbf{v}|| + ||\mathbf{v}||^2 (Cauchy - Schwarzinequality) \\
&\leq (||\mathbf{u}|| + ||\mathbf{v}||)^2 \\
\therefore\ ||\mathbf{u}+\mathbf{v}|| &\leq ||\mathbf{u}|| + ||\mathbf{v}||
\end{aligned}
$$

(The geometric representation of the triangle inequality: for any triangle, the sum of the lengths of any two sides is larger than the length of the third side.)

Length and Dot Product in R^n

Note

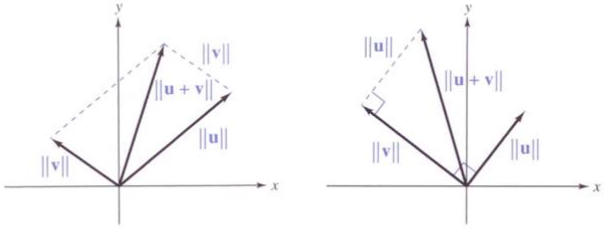
Equality occurs in the triangle inequality if and only if the vectors $\mathbf{u}$ and $\mathbf{v}$ have the same direction (in this situation, $\cos\theta = 1$ and $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}||||\mathbf{v}|| \geq 0$)

Theorem ((5.6) The Pythagorean theorem)

If $\mathbf{u}$ *and* $\mathbf{v}$ *are vector in* R^n*, then* $\mathbf{u}$ *and* $\mathbf{v}$ *are orthogonal if and only if*

$$||\mathbf{u}+\mathbf{v}||^2 = ||\mathbf{u}||^2 + ||\mathbf{v}||^2, \text{(This is because } \mathbf{u}\cdot\mathbf{v} = 0 \text{ in the proof for Theorem 5.5)}$$

(The geometric meaning: for any right triangle , the sum of the squares of the lengths of two legs equals the square of the length of the hypotenuse)



$\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$ $\qquad$ $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$

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Length and Dot Product in R^n

Similarity between dot product and matrix multiplication

$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ (A vector $\mathbf{u} = (u_1, u_2, \ldots, u_n) \in R^n$ can be represented as an $n \times 1$ colum

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} u_1 v_1 + u_2 v_2 + \cdots + u_n v_n \end{bmatrix}$$

(The result of the dot product of $\mathbf{u}$ and $\mathbf{v}$ is the same as the result of the matrix multiplication of $\mathbf{u}^T$ and $\mathbf{v}$)

Inner Product Spaces

Inner product : represented by angle brackets $< \mathbf{u}, \mathbf{v} >$

Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be vectors in a vector space V, and let c be any scalar. An inner product on V is a function that associates a real number with each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ and satisfies the following axioms (abstraction definition from the properties of dot product in Theorem 5.3 on Slide 5.12)

1. $< \mathbf{u}, \mathbf{v} >=< \mathbf{v}, \mathbf{u} >$ (commutative property of the inner product)
2. $< \mathbf{u}, \mathbf{v} + \mathbf{w} >=< \mathbf{v}, \mathbf{u} > + < \mathbf{u}, \mathbf{w} >$ (distributive property of the inner product over vector addition)
3. $c < \mathbf{u}, \mathbf{v} >=< c\mathbf{u}, \mathbf{v} >$ (associative property of the scalar multiplication and the inner product)
4. $< \mathbf{v}, \mathbf{v} >\geq 0$
5. $< \mathbf{v}, \mathbf{v} >= 0$ if and only if $\mathbf{v} = 0$

Note

- $\mathbf{u} \cdot \mathbf{v} =$ dot product (Euclidean inner product for R^n)
- $< \mathbf{u}, \mathbf{v} >=$ general inner product for a vector V

  A vector space V with an inner product is called an inner product space

  Vector space: $(V, +, \cdot)$

  Inner product space: $(V, +, \cdot, <, >)$

The Euclidean inner product for R^n

Example (Note)

Show that the dot product in R^n satisfies the four axioms of an inner product

Solution

$\mathbf{u} = (u_1, u_2, \cdots, u_n), \mathbf{v} = (v_1, v_2, \cdots, v_n)$
$< \mathbf{u}, \mathbf{v} >= \mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$
By Theorem 5.3, this dot product satisfies the required four axioms. Thus, the dot product can be a sort of inner product in R^n

Inner Product Spaces

Example (A different inner product for R^n)

Show that the following function defines an inner product on R^n. Given $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$,

$$< \mathbf{u}, \mathbf{v} >= u_1 v_1 + 2u_2 v_2$$

Solution

1. $< \mathbf{u}, \mathbf{v} >= u_1 v_1 + 2u_2 v_2 = v_1 u_1 + 2v_2 u_2 =< \mathbf{v}, \mathbf{u} >$
2. *Let* $\mathbf{w} = (w_1, w_2)$, *then*

$$\begin{aligned} < \mathbf{u}, \mathbf{v} + \mathbf{w} > &= u_1(v_1 + w_1) + 2u_2(v_2 + w_2) \\ &= u_1 v_1 + u_1 w_1 + 2u_2 v_2 + 2u_2 w_2 = (u_1 v_1 + 2u_2 v_2) + (u_1 w_1 + 2u_2 w_2) \\ &= < \mathbf{u}, \mathbf{v} > + < \mathbf{u}, \mathbf{w} > \end{aligned}$$

3. $c < \mathbf{u}, \mathbf{v} >= c(u_1 v_1 + 2u_2 v_2) = (cu_1)v_1 + 2(cu_2)v_2 =< c\mathbf{u}, \mathbf{v} >$

Solution

4. $< \mathbf{v}, \mathbf{v} >= v_1^2 + 2v_2^2 \geq 0$
5. $< \mathbf{v}, \mathbf{v} >= 0 \implies v_1^2 + 2v_2^2 = 0 \implies v_1 = v_2 = 0(\mathbf{v} = \mathbf{0})$

Note

This example can be generalized such that

$$< \mathbf{u}, \mathbf{v} >= \mathbf{u} \cdot \mathbf{v} = c_1 u_1 v_1 + c_2 u_2 v_2 + \cdots + c_n u_n v_n, \forall c_i > 0$$

can be an inner product on R^n.

Example (A different inner product for R^n)

Show that the following function is not an inner product on R^3. Given $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$,

$$< \mathbf{u}, \mathbf{v} >= u_1 v_1 - 2u_2 v_2 + u_3 v_3$$

Solution

Let $\mathbf{v} = (1, 2, 1)$

Then $< \mathbf{v}, \mathbf{v} >= (1)(1) - 2(2)(2) + (1)(1) = -6 < 0$

Axiom 4 is not satisfied

Thus this function is not an inner product on R^3

Theorem ((5.7) Properties of inner products)

Let $\mathbf{u}, \mathbf{v}$ *and* $\mathbf{w}$ *be vectors in an inner product space* V*, and let* c *be any real number*

1. $< \mathbf{0}, \mathbf{v} >=< \mathbf{v}, \mathbf{0} >= \mathbf{0}$
2. $< \mathbf{u} + \mathbf{v}, \mathbf{w} >=< \mathbf{u}, \mathbf{w} > + < \mathbf{v}, \mathbf{w} >$
3. $< \mathbf{u}, c\mathbf{v} >= c < \mathbf{u}, \mathbf{v} >$

(To prove these properties, you can use only the four axioms for defining an inner product)

Proof.

1. $< \mathbf{0}, \mathbf{v} >=< 0\mathbf{u}, \mathbf{v} > \overset{(3)}{=} 0 < \mathbf{u}, \mathbf{v} >= 0$
2. $< \mathbf{u} + \mathbf{v}, \mathbf{w} > \overset{(1)}{=} < \mathbf{w}, \mathbf{u} + \mathbf{v} > \overset{(2)}{=} < \mathbf{w}, \mathbf{u} > + < \mathbf{w}, \mathbf{v} > \overset{(1)}{=} < \mathbf{u}, \mathbf{w} > + < \mathbf{v}, \mathbf{w} >$
3. $< \mathbf{u}, c\mathbf{v} > \overset{(1)}{=} < c\mathbf{v}, \mathbf{u} > \overset{(3)}{=} c < \mathbf{v}, \mathbf{u} > \overset{(1)}{=} c < \mathbf{u}, \mathbf{v} >$

$\square$

Inner Product Spaces

Dot product in Euclidean n-space

The definition of norm (or length), distance, angle, orthogonal, and normalizing for general inner product spaces closely parallel to those based on the dot product in Euclidean n-space

1. Norm (length) of $\mathbf{u}$:

$$||\mathbf{u}|| = \sqrt{< \mathbf{u}, \mathbf{u} >}$$

2. Distance between $\mathbf{u}$ and $\mathbf{v}$:

$$d(\mathbf{u}, \mathbf{v}) = ||\mathbf{u} - \mathbf{v}|| = \sqrt{< \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} >}$$

3. Angle between two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$:

$$\cos\theta = \frac{< \mathbf{u}, \mathbf{v} >}{||\mathbf{u}||||\mathbf{v}||}$$

4. Orthogonal:
   $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $< \mathbf{v}, \mathbf{v} >= 0$

Normalizing vectors

1. If $||\mathbf{v}|| = 1$, then $\mathbf{v}$ is called a unit vector
   Note that $||\mathbf{v}||$ is defined as $\sqrt{< \mathbf{u}, \mathbf{u} >}$
2. $\mathbf{v} \neq \mathbf{0} \xrightarrow{\textit{Normalizing}} \frac{\mathbf{v}}{||\mathbf{v}||}$ (the unit vector in the direction of $\mathbf{v}$)

Example (An inner product in the polynomial space)

For $p = a_0 + a_1 x + \cdots + a_n x^n$ and $q = b_0 + b_1 x + \cdots + b_n x^n$. Then

$$< p, q >= a_0 b_0 + a_1 b_1 + \cdots + a_n b_n$$

is an inner product space in $_n$.
Let $p(x) = 1 - 2x^2$ and $q(x) = 4 - 2x + x^2$ be polynomials in P_2. Find (1) $< p, q >$ (2) $||q||$ (3) $d(p, q)$

Solution

1. $< p, q >= (1)(4) + (0)(-2) + (-2)(1) = 2$
2. $||q|| = \sqrt{< p, q >} = \sqrt{4^2 + (-2)^2 + 1^2} = \sqrt{21}$
3. $\because p - q = -3 + 2x - 3x^2$
   $\therefore d(p, q) = ||p - q|| = \sqrt{< p - q, p - q >} = \sqrt{(-)^2 + (2)^2 + (-3)^2} = \sqrt{22}$

Inner Product Spaces

Properties of norm: (the same as the properties for the dot product in R^n

1. $||\mathbf{u}|| \geq 0$
2. $||\mathbf{u}|| = 0$ if and only if $\mathbf{u} = \mathbf{0}$
3.
4. $||c\mathbf{u}|| = |c|||\mathbf{u}||$

Properties of distance: (the same as the properties for the dot product in R^n

1. $d(\mathbf{u}, \mathbf{v}) \geq 0$
2. $d(\mathbf{u}, \mathbf{v}) = 0$ if and only if $\mathbf{u} = \mathbf{v}$
3. $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$

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Theorem ((5.8))

Let **u** *and* **v** *be vectors in an inner product space V.*

1. *Cauchy-Schwarz inequality:*

$$| < \mathbf{u}, \mathbf{v} > | \leq ||\mathbf{v}||||\mathbf{v}|| \text{ } \textit{Theorem}5.4$$

2. *Triangle inequality:*

$$||\mathbf{u} + \mathbf{v}|| = ||\mathbf{u}|| + ||\mathbf{v}|| \text{ } \textit{Theorem}5.5$$

3. *Pythagorean theorem:*
   u *and* **v** *are orthogonal if and only if*

$$||\mathbf{u} + \mathbf{v}||^2 = ||\mathbf{u}||^2 + ||\mathbf{v}||^2 \text{ } \textit{Theorem}5.6$$

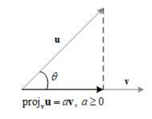
Orthogonal projections

For the dot product function in R^n, we define the orthogonal projection of **u** onto **v** to be $\text{proj}_v\mathbf{u} = a\mathbf{v}$ (a scalar multiple of **u**), and the coefficient a can be derived as follows
Consider $a \geq 0$, then

$$||a\mathbf{v}|| = |a|||\mathbf{v}|| = a||\mathbf{u}||\cos\theta$$

$$= \frac{||\mathbf{u}||||\mathbf{v}||\cos\theta}{||\mathbf{v}||} = \frac{||\mathbf{u}||||\mathbf{v}||}{||\mathbf{v}||}\frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} = \frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{v}||}$$

$$\implies a = \frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{v}||^2} = \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}} \implies \text{porj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v}$$



For inner product spaces:

Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors in an inner product space V. If $\mathbf{v} \neq \mathbf{0}$, then the orthogonal projection of $\mathbf{u}$ onto $\mathbf{v}$ is given by

$$\text{porj}_{\mathbf{v}}\mathbf{u} = \frac{< \mathbf{u}, \mathbf{v} >}{< \mathbf{v}, \mathbf{v} >}\mathbf{v}$$

Example (Finding an orthogonal projection in R^3)

Use the Euclidean inner product in R^3 to find the orthogonal projection of $\mathbf{u} = (6, 2, 4)$ onto $\mathbf{v} = (1, 2, 0)$

Solution

$\because < \mathbf{u}, \mathbf{u} >= (6)(1) + (2)(2) + (4)(0) = 10$

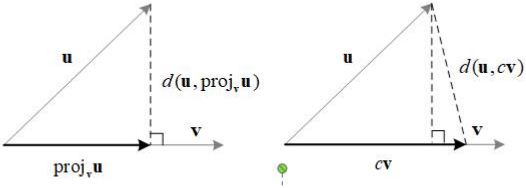
$< \mathbf{v}, \mathbf{v} >= 1^2 + 2^2 + 0^2 = 5$

$\therefore \; porj_{\mathbf{v}}\mathbf{u} = \frac{<\mathbf{u},\mathbf{v}>}{<\mathbf{v},\mathbf{v}>}\mathbf{v} = \frac{10}{5}(1, 2, 0) = (2, 4, 0)$

Theorem ((5.9)Orthogonal projection and distance)

Let $\mathbf{u}$ *and* $\mathbf{v}$ *be vectors in an inner product space* V *and if* $\mathbf{v} \neq \mathbf{0}$, *then*

$$d(\mathbf{u},) < d(\mathbf{u}, c\mathbf{v}), c \neq \frac{<\mathbf{u},\mathbf{u}>}{<\mathbf{v},\mathbf{v}>}$$



Theorem 5.9 can be inferred straightforwardly by the Pythagorean Theorem, i.e., in a right triangle, the hypotenuse is longer than both legs

Orthonormal Bases: Gram-Schmidt Process

Orthonormal Bases: Gram-Schmidt Process

Orthogonal set

A set S of vectors in an inner product space V is called an orthogonal set if every pair of vectors in the set is orthogonal

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\} \subseteq V$$
$$< \mathbf{v}_i, \mathbf{v}_j >= 0, \forall i \neq j$$

Orthonormal set

An orthogonal set in which each vector is a unit vector is called orthonormal set

$$\begin{cases} \text{For } i = j, < \mathbf{v}_i, \mathbf{v}_j >=< \mathbf{v}_i, \mathbf{v}_i >= ||\mathbf{v}_i||^2 = 1 \\ \text{For } i \neq j, < \mathbf{v}_i, \mathbf{v}_j >= 0 \end{cases}$$

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Note

- If S is also a basis, then it is called an orthogonal basis or an orthonormal basis
- The standard basis for R^n is orthonormal. For example,

$$S = \{(1,0,0),(0,1,0),(0,0,1)\}$$

is an orthonormal basis for R^3

This section identifies some advantages of orthonormal bases, and develops a procedure for constructing such bases, known as Gram-Schmidt orthonormalization process

Orthonormal Bases: Gram-Schmidt Process

Example (A nonstandard orthonormal basis for R^3)

Show that the following set is an orthonormal basis

$$S = \left\{ \overbrace{\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)}^{\mathbf{v}_1} \overbrace{\left(-\frac{\sqrt{2}}{6}, \frac{\sqrt{2}}{6}, \frac{2\sqrt{2}}{3}\right)}^{\mathbf{v}_2} \overbrace{\left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}\right)}^{\mathbf{v}_3} \right\}$$

Orthonormal Bases: Gram-Schmidt Process

Solution

- *First, show that the three vectors are mutually orthogonal*

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = -\frac{1}{6} + \frac{1}{6} + 0 = 0$$
$$\mathbf{v}_1 \cdot \mathbf{v}_3 = \frac{2}{3\sqrt{2}} - \frac{2}{3\sqrt{2}} + 0 = 0$$
$$\mathbf{v}_2 \cdot \mathbf{v}_3 = -\frac{\sqrt{2}}{9} - \frac{\sqrt{2}}{9} + -\frac{2\sqrt{2}}{9} = 0$$

Thus S is an orthogonal set

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Solution

- *Second, show that each vector is of length 1*

$$||\mathbf{v}_1|| = \sqrt{\mathbf{v}_1 \cdot \mathbf{v}_1} = \sqrt{\frac{1}{2} + \frac{1}{2} + 0} = 1$$

$$||\mathbf{v}_2|| = \sqrt{\mathbf{v}_2 \cdot \mathbf{v}_2} = \sqrt{\frac{2}{36} + \frac{2}{26} + \frac{8}{9}} = 1$$

$$||\mathbf{v}_3|| = \sqrt{\mathbf{v}_3 \cdot \mathbf{v}_3} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9}} = 1$$

Thus S is an orthonormal set

Orthonormal Bases: Gram-Schmidt Process

Solution

- *Thirdly, show that three vectors are linearly independent*
  Because these three vectors are linearly independent (you can check by solving $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = 0$*) in* R^3 *(of dimension 3), by Theorem 4.12 (given a vector space with dimension n, then n linearly independent vectors can form a basis for this vector space), these three linearly independent vectors form a basis for* R^3.

$\Longrightarrow$ *S is a (nonstandard) orthonormal basis for* R^3

Example (An orthonormal basis for $P_2(x)$)

In $P_2(x)$, with the inner product $< p, q >= a_0 b_0 + a_1 b_1 + a_2 b_2$, the standard basis $B = \{1, x, x^2\}$ is orthonormal

Solution

- *First, identify the vectors and show that they are mutually orthogonal*

$$\mathbf{v}_1 = 1 + 0x + 0x^2, \qquad \mathbf{v}_2 = 0 + x + 0x^2, \qquad \mathbf{v}_3 = 0 + 0x + x^2, \textit{ then}$$

$$\begin{aligned}
\mathbf{v}_1 \cdot \mathbf{v}_2 &= (1)(0) + (0)(1) + (0)(0) = 0 \\
\mathbf{v}_1 \cdot \mathbf{v}_3 &= (1)(0) + (0)(0) + (0)(1) = 0 \\
\mathbf{v}_2 \cdot \mathbf{v}_3 &= (0)(0) + (1)(0) + (0)(1) = 0
\end{aligned}$$

Thus B is an orthogonal set

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Solution

- *Second, show that each vector is of length 1*

$$||\mathbf{v}_1|| = \sqrt{\mathbf{v}_1 \cdot \mathbf{v}_1} = \sqrt{(1)(1) + (0)(0) + (0)(0)} = 1$$
$$||\mathbf{v}_2|| = \sqrt{\mathbf{v}_2 \cdot \mathbf{v}_2} = \sqrt{(0)(0) + (1)(1) + (0)(0)} = 1$$
$$||\mathbf{v}_3|| = \sqrt{\mathbf{v}_3 \cdot \mathbf{v}_3} = \sqrt{(0)(0) + (0)(0) + (1)(1)} = 1$$

Thus B is an orthonormal set

Orthonormal Bases: Gram-Schmidt Process

Solution

- *Thirdly, show that three vectors are linearly independent*
  Because these three vectors are linearly independent (you can check by solving $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = 0$*) in* $P_2(x)$ *(of dimension 3), by Theorem 4.12 (given a vector space with dimension* n*, then* n *linearly independent vectors can form a basis for this vector space), these three linearly independent vectors form a basis for* $P_2(x)$*.*

$\Longrightarrow$ *S is a (nonstandard) orthonormal basis for* R^3

Theorem ((5.10): Orthogonal sets are linearly independent)

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthogonal set of nonzero vectors in an inner product space V, then S is linearly independent

Proof.

S is an orthogonal set of nonzero vectors, that is,

$$< \mathbf{v}_i, \mathbf{v}_j >= 0, \forall i \neq j, \text{ and } < \mathbf{v}_i, \mathbf{v}_j >> 0$$

$$\begin{aligned}
\text{For } c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n &= \mathbf{0} \\
\Longrightarrow < c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n, \mathbf{v}_i > &= < \mathbf{0}, \mathbf{v}_i >= \mathbf{0}\,\forall i \\
\Longrightarrow c_1 < \mathbf{v}_1, \mathbf{v}_i > + c_2 < \mathbf{v}_2, \mathbf{v}_i > + \cdots + c_n < \mathbf{v}_n, \mathbf{v}_i > = c_i < \mathbf{v}_i, \mathbf{v}_i > &= \mathbf{0} \\
\because < \mathbf{v}_i, \mathbf{v}_i > \neq 0 \Longrightarrow c_i &= 0, \forall i \\
\therefore S \text{ is linearly independent}
\end{aligned}$$

Orthonormal Bases: Gram-Schmidt Process

Corollary to Theorem 5.10

If V is an inner product space with dimension n, then any orthogonal set of n nonzero vectors is a basis for V

- By Theorem 5.10, if $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthogonal set of n vectors, then S is linearly independent.
- According to Theorem 4.12, if $S = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a linearly independent set of n vectors in V (with dimension n), then S is a basis for V

Based on the above two arguments, it is straightforward to derive the above corollary to Theorem 5.10

Orthonormal Bases: Gram-Schmidt Process

Example (Using orthogonality to test for a basis)

Show that the following set is a basis for R^4

$$S = \left\{ \overset{\mathbf{v}_1}{(2,3,2,-2)}, \overset{\mathbf{v}_2}{(1,0,0,1)}, \overset{\mathbf{v}_3}{(-1,0,2,1)}, \overset{\mathbf{v}_4}{(-1,2,-1,1)} \right\}$$

Orthonormal Bases: Gram-Schmidt Process

Solution

$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ *are nonzero vectors*

$$\begin{aligned}
\mathbf{v}_1 \cdot \mathbf{v}_2 &= 2+0+0-2=0 & \mathbf{v}_2 \cdot \mathbf{v}_3 &= -1+0+0+1=0 \\
\mathbf{v}_1 \cdot \mathbf{v}_3 &= -2+0+4+-2=0 & \mathbf{v}_2 \cdot \mathbf{v}_4 &= -1+0+4++1=0 \\
\mathbf{v}_1 \cdot \mathbf{v}_4 &= -2+6-2-2=0 & \mathbf{v}_3 \cdot \mathbf{v}_4 &= 1+0-2+1=0
\end{aligned}$$

$\Longrightarrow$ *S is orthogonal*

$\Longrightarrow$ *S is a basis for* R^4 *(by Corollary to Theorem 5.10)*

(The corollary to Thm. 5.10 shows an advantage of introducing the concept of orthogonal vectors, i.e., it is not necessary to solve linear systems to test whether S is a basis if S is a set of orthogonal vectors)

Orthonormal Bases: Gram-Schmidt Process

Theorem ((5.11): Coordinates relative to an orthonormal basis)

If $B = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthonormal basis for an inner product space V, then the unique coordinate representation of a vector $\mathbf{w}$ with respect to B is

$$\mathbf{w} = < \mathbf{w}, \mathbf{v}_1 > \mathbf{v}_1 + < \mathbf{w}, \mathbf{v}_2 > \mathbf{v}_2 + \cdots + < \mathbf{w}, \mathbf{v}_n > \mathbf{v}_n$$

The above theorem tells us that it is easy to derive the coordinate representation of a vector relative to an orthonormal basis, which is another advantage of using orthonormal bases

Orthonormal Bases: Gram-Schmidt Process

Proof.

$B = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthonormal basis for V

$$\mathbf{w} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_n\mathbf{v}_n \in V$$

Since $< \mathbf{v}_i, \mathbf{v}_j >= \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$, then

$$\begin{aligned} < \mathbf{w}, \mathbf{v}_i >= & < (k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_n\mathbf{v}_n), \mathbf{v}_i > \\ = & k_1 < \mathbf{v}_1, \mathbf{v}_i > + k_2 < \mathbf{v}_2, \mathbf{v}_i > + \cdots + k_n < \mathbf{v}_n, \mathbf{v}_i > \\ = & k_i, \forall i = 1, 2, \ldots, n \end{aligned}$$

$$\implies \mathbf{w} = < \mathbf{w}, \mathbf{v}_1 > \mathbf{v}_1 + < \mathbf{w}, \mathbf{v}_2 > \mathbf{v}_2 + \cdots + < \mathbf{w}, \mathbf{v}_n > \mathbf{v}_n$$

$\square$

Orthonormal Bases: Gram-Schmidt Process

Note

If $B = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthonormal basis for V and $\mathbf{w} \in V$, Then the corresponding coordinate matrix of $\mathbf{w}$ relative to B is

$$[\mathbf{w}]_B = \begin{bmatrix} < \mathbf{w}, \mathbf{v}_1 > \\ < \mathbf{w}, \mathbf{v}_2 > \\ \vdots \\ < \mathbf{w}, \mathbf{v}_n > \end{bmatrix}$$

Orthonormal Bases: Gram-Schmidt Process

Example (Representing vectors relative to standard basis)

For $\mathbf{w} = (5, -5, 2)$, find its coordinates relative to the standard basis for R^3.

$$< \mathbf{w}, \mathbf{v}_1 >= \mathbf{w} \cdot \mathbf{v}_1 = (5, -5, 2) \cdot (1, 0, 0) = 5$$
$$< \mathbf{w}, \mathbf{v}_2 >= \mathbf{w} \cdot \mathbf{v}_2 = (5, -5, 2) \cdot (0, 1, 0) = -5$$
$$< \mathbf{w}, \mathbf{v}_3 >= \mathbf{w} \cdot \mathbf{v}_3 = (5, -5, 2) \cdot (0, 0, 1) = 2$$
$$\Longrightarrow [\mathbf{w}]_B = \begin{bmatrix} 5 \\ -5 \\ 2 \end{bmatrix}$$

- In fact, it is not necessary to use Thm. 5.11 to find the coordinates relative to the standard basis, because we know that the coordinates of a vector relative to the standard basis are the same as the components of that vector

Orthonormal Bases: Gram-Schmidt Process

- The advantage of the orthonormal basis emerges when we try to find the coordinate matrix of a vector relative to an nonstandard orthonormal basis

Example (Representing vectors relative to an orthonormal basis)

For $\mathbf{w} = (5, -5, 2)$, find its coordinates relative to the orthonormal basis for R^3.

$$B = \left\{ \overset{\mathbf{v}_1}{\left(\frac{3}{5}, \frac{4}{5}, 0\right)}, \overset{\mathbf{v}_2}{\left(-\frac{4}{5}, \frac{3}{5}, 0\right)}, \overset{\mathbf{v}_3}{(0, 0, 1)} \right\}$$

Solution

$$< \mathbf{w}, \mathbf{v}_1 >= \mathbf{w} \cdot \mathbf{v}_1 = (5, -5, 2) \cdot (\frac{3}{5}, \frac{4}{5}, 0) = -1$$

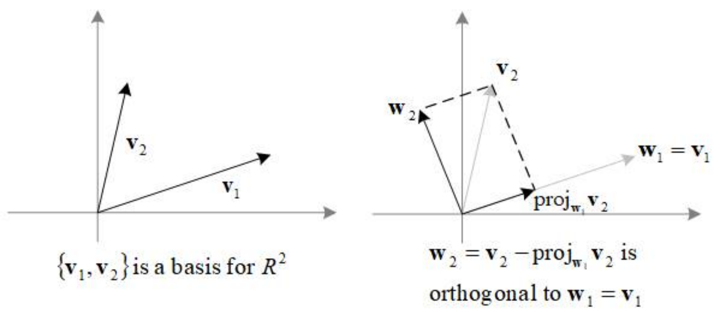
$$< \mathbf{w}, \mathbf{v}_2 >= \mathbf{w} \cdot \mathbf{v}_2 = (5, -5, 2) \cdot (-\frac{4}{5}, \frac{3}{5}, 0) = -7$$

$$< \mathbf{w}, \mathbf{v}_3 >= \mathbf{w} \cdot \mathbf{v}_3 = (5, -5, 2) \cdot (0, 0, 1) = 2$$

$$\implies [\mathbf{w}]_B = \begin{bmatrix} -1 \\ -7 \\ 2 \end{bmatrix}$$

Orthonormal Bases: Gram-Schmidt Process

The geometric intuition of the Gram-Schmidt process to find an orthonormal basis in R^2



$$\implies \left\{ \frac{\mathbf{w}_1}{||\mathbf{w}_1||}, \frac{\mathbf{w}_2}{||\mathbf{w}_2||} \right\} \text{ is an orthonormal basis in } R^2$$

Orthonormal Bases: Gram-Schmidt Process

Gram-Schmidt orthonormalization process

$B = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthonormal basis for V

$$\begin{aligned}
\text{Let } \mathbf{w}_1 =& \mathbf{v}_1 \\
\mathbf{w}_2 =& \mathbf{v}_2 - \text{proj}_{S_1}\mathbf{v}_2 = \mathbf{v}_2 - \frac{<\mathbf{v}_2, \mathbf{w}_1>}{<\mathbf{w}_1, \mathbf{w}_1>}\mathbf{w}_1, \quad S_1 = \text{span}(\mathbf{w}_1) \\
\mathbf{w}_3 =& \mathbf{v}_3 - \text{proj}_{S_2}\mathbf{v}_3 = \mathbf{v}_3 - \frac{<\mathbf{v}_3, \mathbf{w}_1>}{<\mathbf{w}_1, \mathbf{w}_1>}\mathbf{w}_1 - \frac{<\mathbf{v}_3, \mathbf{w}_2>}{<\mathbf{w}_2, \mathbf{w}_2>}\mathbf{w}_2, \quad S_2 = \text{span}(\mathbf{w}_1, \mathbf{w}_2) \\
&\vdots \\
\mathbf{w}_n =& \mathbf{v}_n - \text{proj}_{S_{n-1}}\mathbf{v}_n = \mathbf{v}_n - \sum_{i=1}^{n-1}\frac{<\mathbf{v}_n, \mathbf{w}_i>}{<\mathbf{w}_i, \mathbf{w}_i>}\mathbf{w}_i, \quad S_{n-1} = \text{span}(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_{n-1})
\end{aligned}$$

Orthonormal Bases: Gram-Schmidt Process

Gram-Schmidt orthonormalization process

$$\implies B^{'} = \{\mathbf{w}_1, \mathbf{w}_2, \cdots, \mathbf{w}_n\} \text{ is an orthogonal basis}$$

$$\implies B^{''} = \left\{\frac{\mathbf{w}_1}{||\mathbf{w}_1||}, \frac{\mathbf{w}_2}{||\mathbf{w}_2||}, \cdots, \frac{\mathbf{w}_n}{||\mathbf{w}_n||}\right\} \text{ is an orthonormal basis}\}$$

Example (Applying the Gram-Schmidt orthonormalization process)

Apply the Gram-Schmidt process to the following basis for R^3

$$B = \left\{ (1, \overset{\mathbf{v}_1}{1}, 0), (1, \overset{\mathbf{v}_2}{2}, 0), (0, \overset{\mathbf{v}_3}{1}, 2) \right\}$$

Orthonormal Bases: Gram-Schmidt Process

Solution

$$
\begin{aligned}
\mathbf{w}_1 &= \mathbf{v}_1 = (1, 1, 0) \\
\mathbf{w}_2 &= \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1}\mathbf{w}_1 = (1, 2, 0) - \frac{3}{2}(1, 1, 0) = (-\frac{1}{2}, \frac{1}{2}, 0) \\
\mathbf{w}_3 &= \mathbf{v}_3 - \frac{\mathbf{v}_3 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1}\mathbf{w}_1 - \frac{\mathbf{v}_3 \cdot \mathbf{w}_2}{\mathbf{w}_2 \cdot \mathbf{w}_2}\mathbf{w}_2 = (0, 1, 2) - \frac{1}{2}(1, 1, 0) - \frac{1/2}{1/2}(-\frac{1}{2}, \frac{1}{2}, 0) = (0, 0, 2)
\end{aligned}
$$

Orthogonal basis

$\Longrightarrow \; B' = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} = \left\{(1, 1, 0), (-\frac{1}{2}, \frac{1}{2}, 0), (0, 0, 2)\right\}$

Orthonormal basis

$\Longrightarrow \; B'' = \left\{\frac{\mathbf{w}_1}{||\mathbf{w}_1||}, \frac{\mathbf{w}_2}{||\mathbf{w}_2||}, \frac{\mathbf{w}_3}{||\mathbf{w}_3||}\right\} = \left\{(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0), (\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0), (0, 0, 1)\right\}$

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Example (Alternative form of Gram-Schmidt orthonormalization process)

Find an orthonormal basis for the solution space of the homogeneous system of linear equations:

$$\begin{array}{rcrcrcrcr} x_1 & + & x_2 & & & + & 7x_4 & = & 0 \\ 2x_1 & + & x_2 & + & 2x_3 & + & 6x_4 & = & 0 \end{array}$$

Orthonormal Bases: Gram-Schmidt Process

Solution

$$\begin{bmatrix} 1 & 1 & 0 & 7 & 0 \\ 2 & 1 & 2 & 6 & 0 \end{bmatrix} \xrightarrow{G.J.E} \begin{bmatrix} 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & -2 & 8 & 0 \end{bmatrix}$$

$$\Longrightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -2s+t \\ 2s-8t \\ s \\ t \end{bmatrix} = s\begin{bmatrix} -2 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} 1 \\ -8 \\ 0 \\ 1 \end{bmatrix}$$

Thus one basis for the solution space is

$$B = \Big\{ \overset{\mathbf{v}_1}{(-2, 2, 1, 0)}, \overset{\mathbf{v}_2}{(1, -8, 0, 1)} \Big\}$$

Orthonormal Bases: Gram-Schmidt Process

Solution

$$\mathbf{w}_1 = \mathbf{v}_1 = (-2, 2, 1, 0) \textit{ and } \mathbf{u}_1 = \frac{\mathbf{w}_1}{||\mathbf{w}_1||} = \frac{1}{3}(-2, 2, 1, 0) = \left(\frac{-2}{3}, \frac{2}{3}, \frac{1}{3}, 0\right)$$

$$\mathbf{w}_2 = \mathbf{v}_2 - < \mathbf{v}_2, \mathbf{u}_1 > \mathbf{u}_1 \text{ (} \textit{due to } \mathbf{w}_2 = \mathbf{v}_2 - \frac{< \mathbf{v}_2, \mathbf{u}_1 >}{< \mathbf{u}_1, \mathbf{u}_1 >}\mathbf{u}_1 \textit{ and } < \mathbf{u}_1, \mathbf{u}_1 >= 1)$$

$$= (1, -8, 0, 1) - \left[(1, -8, 0, 1) \cdot \left(\frac{-2}{3}, \frac{2}{3}, \frac{1}{3}, 0\right)\right]\left(\frac{-2}{3}, \frac{2}{3}, \frac{1}{3}, 0\right)$$

$$= (-3, -4, 2, 1)$$

$$\mathbf{u}_2 = \frac{\mathbf{w}_2}{||\mathbf{w}_2||} = \frac{1}{\sqrt{30}}(-3, -4, 2, 1)$$

Orthonormal basis

$$\implies B'' = \{\mathbf{u}_1, \mathbf{u}_2\} = \left\{\left(\frac{-2}{3}, \frac{2}{3}, \frac{1}{3}, 0\right), \left(\frac{-3}{\sqrt{30}}, \frac{-4}{\sqrt{30}}, \frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}\right)\right\}$$

Orthonormal Bases: Gram-Schmidt Process

Alternative form of the Gram-Schmidt orthogonalization process

$B = \{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is an orthonormal basis for V

$$\mathbf{u}_1 = \frac{\mathbf{w}_1}{||\mathbf{w}_1||} = \frac{\mathbf{v}_1}{||\mathbf{v}_1||}$$

$$\mathbf{u}_2 = \frac{\mathbf{w}_2}{||\mathbf{w}_2||}, \text{ where } \mathbf{w}_2 = \mathbf{v}_2 - < \mathbf{v}_2, \mathbf{u}_1 > \mathbf{u}_1$$

$$\mathbf{u}_3 = \frac{\mathbf{w}_3}{||\mathbf{w}_3||}, \text{ where } \mathbf{w}_3 = \mathbf{v}_3 - < \mathbf{v}_3, \mathbf{u}_1 > \mathbf{u}_1 - < \mathbf{v}_3, \mathbf{u}_2 > \mathbf{u}_2$$

$$\vdots$$

$$\mathbf{u}_n = \frac{\mathbf{w}_n}{||\mathbf{w}_n||}, \text{ where } \mathbf{w}_n = \mathbf{v}_n - \sum_{i=1}^{n-1} < \mathbf{v}_n, \mathbf{u}_i > \mathbf{u}_i$$

$\implies \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is an orthonormal basis for V